ELSTE Data science

Class 2: Exploratory data analysis

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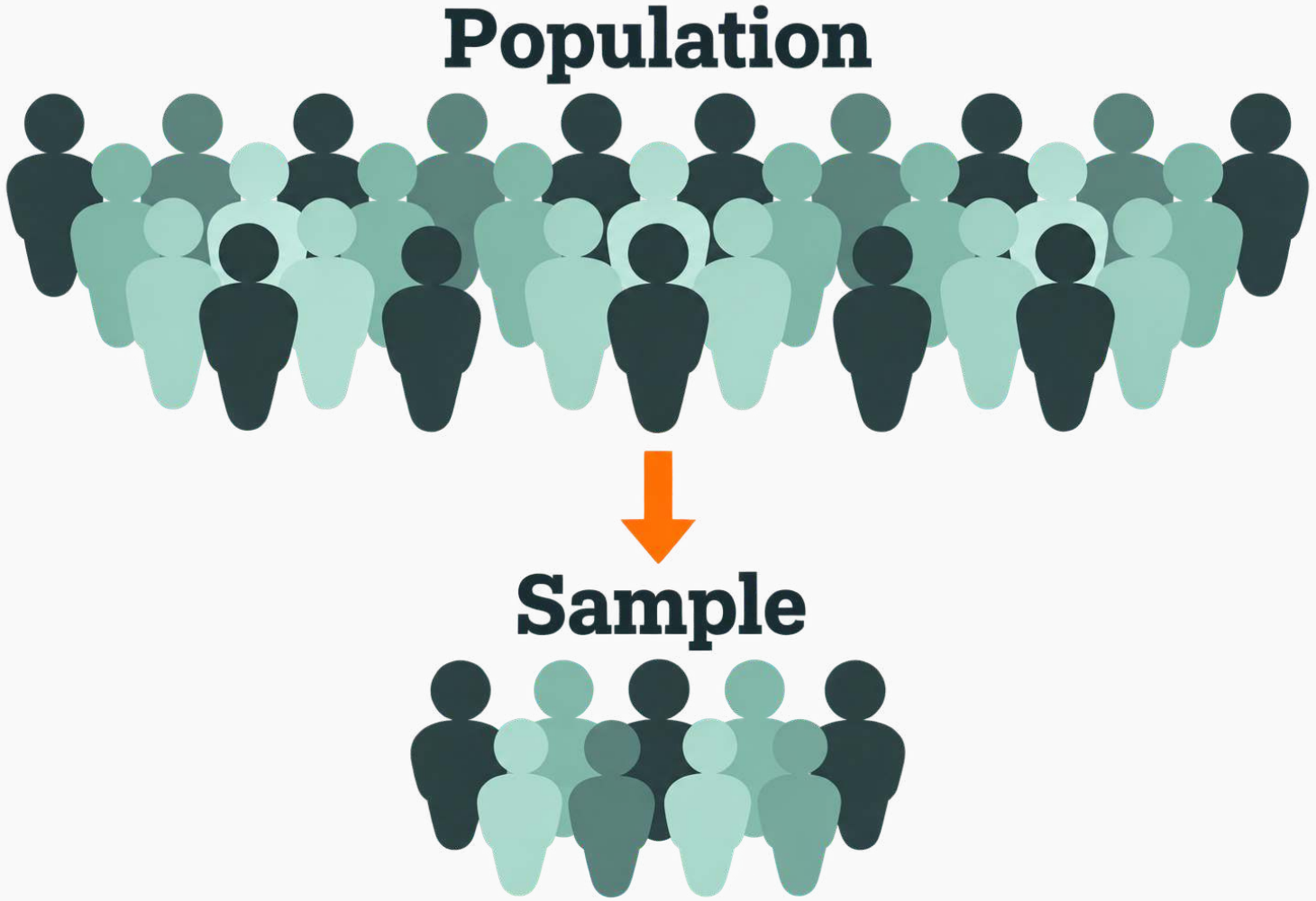
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Today's objectives

- **Last week**: We learned how to handle data using `Pandas`
  - → Load, access, query, filter, sort and operate on data

- **This week:** You received / compiled / produced a new dataset
- Review common tasks / method to get accustomed to the dataset
  - → Understand its **structure** → what type of data does it contain
  - → Explore it **visually** → plotting with dedicated libraries
  - → Describe **single variables** → univariate analyses
  - → Assess coupled behaviours of **two variables** → bivariate analyses
  - → First attempt at **modelling** these behaviours → linear regressions

Sample vs population



The dataset

Case study

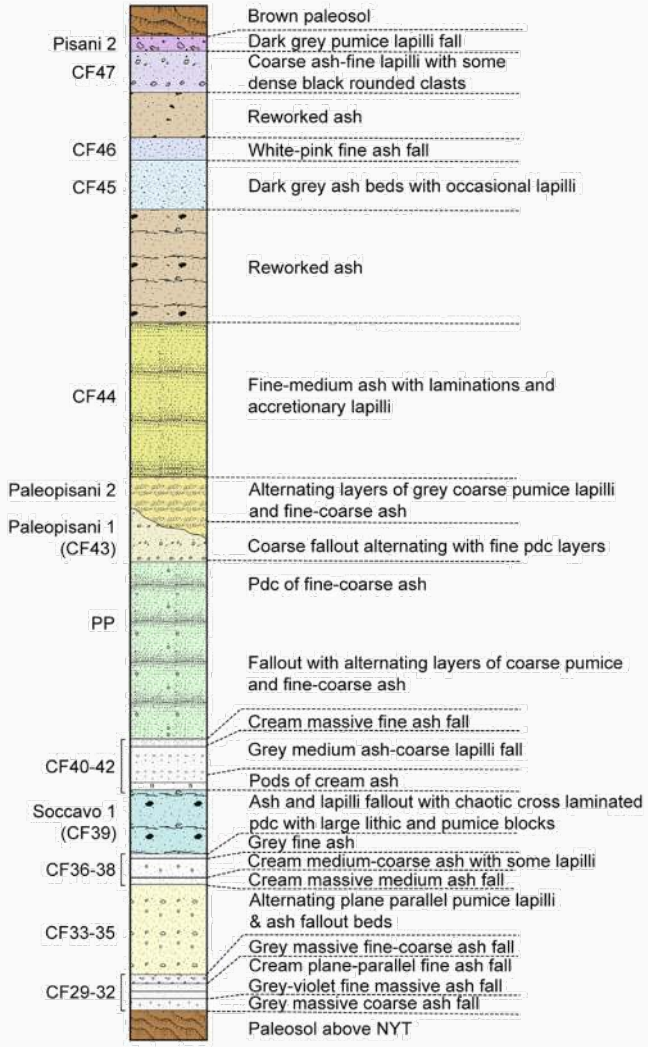
- **Campi Flegrei**: Home to 1.5 million people
- **Caldera-forming** eruptions
  - → Campanian Ignimbrite (CI), ~39 ka ago
  - → Neapolitan Yellow Tuff (NYT), ~15 ka ago



Dataset

Tephrostratigraphy and glass compositions of post-15 kyr Campi Flegrei eruptions

- Smith et al. (2011)
- Major and trace elements



Loading the dataset

- Import `pandas`
- Load an **excel** file using `pandas`
- Specify **which sheet** to read from using the `sheet_name` argument
  - → *Supp_majors* → `df_majors`
  - → *Supp_traces* → `df_traces`

```
1 # Load Pandas
2 import pandas as pd # Import pandas
3
4 # Import the dataset specifying which Excel sheet name to load the data from
5 df_majors = pd.read_csv('https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data
6 df_traces = pd.read_csv('https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data
```

Inspecting the dataset

```
1 df_traces.head(5)
```

	Analysis no.	Strat. Pos.	Eruption	controlcode	Sa
0	1915	63	Astroni 7	1	79
1	1916	63	Astroni 7	1	79
2	1917	63	Astroni 7	1	79
3	1918	63	Astroni 7	1	79
4	1919	63	Astroni 7	1	79

5 rows × 37 columns

```
1 df_traces.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 370 entries, 0 to 369
Data columns (total 37 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Analysis no.      370 non-null    int64
 1   Strat. Pos.       370 non-null    int64
 2   Eruption          370 non-null    object
 3   controlcode       370 non-null    int64
 4   Sample            370 non-null    object
 5   Epoch             370 non-null    object
 6   Crater size       370 non-null    int64
 7   Date of analysis  370 non-null    object
 8   Si/bulk cps       370 non-null
float64
 9   SiO2* (EMP)       370 non-null
float64
 10  Sc                370 non-null
```

Types of data

Basic types of data

- `int`: **Integer** numbers → 0, 1, 2, …
- `float`: **Decimal** numbers → 1.1, 1.2, 1.3, …
- `object`: **Strings** → “Campi Flegrei”

A note on *Bytes*

- `int` and `float` numbers are followed by 8, 16, 32 or 64 → **Byte** → control how much memory a variable uses
- A **Bit** is a binary digit → 0/1 → smallest possible unit of data
- `int8` can store 2^8 digits = 256
- `int16` can store 2^16 digits = 65’563

Families of data

Numerical data

- Represent quantities or measurable values
- **Quantitative**
  - → *Discrete*: Earthquake counts
  - → *Continuous*: Temperature
- Stored as **numerical** values
- **Operations** → arithmetic

Categorical data

- Represent categories or groups of data
- **Qualitative** / semi-quantitative
  - → *Nominal*: No order → Landslide type
  - → *Ordinal*: Order → rank
- Stored as **strings** or **integer**
- **Operations** → counting, grouping

Families of data - cheat sheet

Feature	Categorical Data	Numerical Data
Definition	**Categories or groups** of data	**Quantities or measurable values**
Nature	Qualitative (describes qualities)	Quantitative (describes amounts or measurements)
Data Type	Non-numeric (often text or labels)	Numeric (numbers only)
Examples	Eruption type	Element concentration
Possible Operations	Counting, grouping, mode	Arithmetic operations
Measurement Scale	Nominal or Ordinal	Interval or Ratio
Visualization Tools	Bar chart, Pie chart	Histogram, Box plot, Scatter plot
Subtypes	**Nominal:** Categories with no order (e.g., color); **Ordinal:** Categories with order (e.g., rank)	**Discrete:** Countable numbers (e.g., number of earthquakes); **Continuous:** Measurable values (e.g., temperature)
Examples of Statistical Summary	Frequency, mode	Mean, median, standard deviation
Storage Format	Strings or labels	Integers or floats

Today

Exploratory data analysis

> *Can fractional crystallisation explain the chemical variability of Campi Flegrei glasses, or do we need magma mixing?*

- **Univariate**: Are element concentrations narrowly distributed, broadly spread or multimodal?
- **Bivariate**: Are relationships such as Sr–Zr consistent with Sr removal by feldspar crystallisation while Zr becomes enriched in the residual melt?
- **Interpretation**: Can one differentiation process explain the observations, or are multiple magma components needed?

  → **Correlation ≠ mechanistic explanation**
- **Sample vs population**: Do the analysed glasses capture the compositional variability within each eruption?

Objectives

1. Understand **data structure**
    - Type and completeness of dataset
2. Basics of **plotting** with `matplotlib` and `seaborn`
3. **Univariate** statistics
    - Position, spread, shape
    - Quantification + visualisation
4. **Bivariate** statistics
    - Covariance & Correlation
    - Intro to regression + residual analysis

Your turn!

- Go to https://elste-master.github.io/Data-Science/
  - → Class 2
  - → Explore the structure of the glass geochemistry or your dataset
  - → Get used to the various functions

